

Maximum Likelihood Estimation

- The **likelihood** is the probability of the observed data given the parameters:

$$P(x, y; \theta)$$

x : the set of samples
 y : their corresponding labels
 θ : model parameters

- In discriminative models we often talk about the **conditional likelihood**:

$$P(y|x; \theta)$$

- The maximum estimator (MLE) is defined as $\theta_{MLE} = \operatorname{argmax}_{\theta} P(x, y; \theta)$
- And in discriminative models as $\theta_{MLE} = \operatorname{argmax}_{\theta} P(y|x; \theta)$

Maximum Likelihood Estimation

- Maximum likelihood estimation is a general estimation approach, commonly used in statistics
- In essence, it seeks to find the parameters under which the observed data will be the most likely